

EXPT NO:4	Visual encoding of data
DATE: 20.01.2026	

PRE-LAB QUESTIONS

1. How does the human visual system process visual variables?

The human visual system processes visual variables such as color, size, shape, and position by rapidly identifying patterns, contrasts, and differences. This enables users to understand complex information quickly, with particular efficiency in perceiving spatial relationships and variations in color and size.

2. Why is improper encoding misleading in analytics?

Improper encoding can mislead analysis by distorting, hiding, or overstating data patterns. Inappropriate use of visual variables may cause viewers to misinterpret trends, comparisons, or relative importance, resulting in flawed conclusions and decisions.

3. What are perceptual limitations in visualization?

Perceptual limitations include challenges in distinguishing subtle color differences, small variations in size, and complex shapes—especially when multiple variables are presented simultaneously. Excessive visual complexity can also lead to cognitive overload and reduced comprehension.

4. How does color choice affect interpretation?

Color selection plays a critical role in interpretation, as colors affect attention, readability, and emotional response. Poor contrast or unsuitable color schemes can confuse users and obscure important insights.

5. Why should AI dashboards be perceptually optimized?

Perceptual optimization ensures that AI dashboards communicate information clearly, accurately, and efficiently. Well-designed visuals reduce cognitive effort, minimize interpretation errors, and support effective human-AI decision-making.

OBJECTIVE: To apply effective visual encoding principles for meaningful data communication.

SCENARIO: A city traffic department analyzes accident data to identify high-risk zones and accident severity patterns.

IN-LAB TASKS (Using R Language) • Encode severity using color gradients • Represent accident frequency using size • Use shape to indicate accident type

CODE:

```
CODE4R*
Source on Save
Run

1 library(ggplot2)
2 library(dplyr)
3
4 data <- read.csv("D:/1/COLLAGE/AI&DS/SEM 6/EDA/LAB/E4/4.traffic_accidents.csv")
5 print("Name: Santhiya Joe Harson J , Roll no : 23BAD102")
6 plot_data <- data %>%
7   group_by(Location, Accident_Type, Severity) %>%
8   summarise(Frequency = n(), .groups = 'drop')
9
10 plot_data$Severity <- factor(plot_data$Severity, levels = c("Minor", "Major", "Fatal"))
11
12
13 set.seed(123)
14
15 ggplot(plot_data, aes(x = Location, y = Severity)) +
16   geom_jitter(aes(size = Frequency,
17                 color = Severity,
18                 shape = Accident_Type,
19                 width = 0.2, height = 0.1, alpha = 0.8)) +
20   scale_color_manual(values = c("Minor" = "#006D2C",
21                               "Major" = "#31A354",
22                               "Fatal" = "#A1D99B")) +
23   scale_size_continuous(range = c(5, 15)) +
24
25   labs(title = "Traffic Accident Analysis: High-Risk Zones",
26        subtitle = "Visualizing Severity, Frequency, and Accident Type",
27        x = "Location Zone",
28        y = "Accident Severity",
29        size = "Accident\nFrequency",
30        shape = "Accident\nType",
31        color = "Severity\nLevel") +
32
33   theme_minimal() +
34   theme(
35     plot.title = element_text(face = "bold", size = 16),
36     axis.text = element_text(size = 12),
37     legend.position = "right",
38     legend.key.size = unit(0.4, "cm"),
39     legend.text = element_text(size = 8),
40     legend.title = element_text(size = 9, face = "bold"),
41     legend.spacing.y = unit(0.2, "cm")
42   )
43
44 }
```

OUTPUT:

```
Source
R 4.5.2 · D:/1/COLLAGE/AI&DS/SEM 6/EDA/LAB/E4/
> library(ggplot2)
> library(dplyr)
>
> data <- read.csv("D:/1/COLLAGE/AI&DS/SEM 6/EDA/LAB/E4/4.traffic_accidents.csv")
> print("Name: Santhiya Joe Harson J , Roll no : 23BAD102")
[1] "Name: Santhiya Joe Harson J , Roll no : 23BAD102"
> plot_data <- data %>%
+   group_by(Location, Accident_Type, Severity) %>%
+   summarise(Frequency = n(), .groups = 'drop')
>
> plot_data$Severity <- factor(plot_data$Severity, levels = c("Minor", "Major", "Fatal"))
>
> set.seed(123)
>
> ggplot(plot_data, aes(x = Location, y = Severity)) +
+   geom_jitter(aes(size = Frequency,
+                   color = Severity,
+                   shape = Accident_Type,
+                   width = 0.2, height = 0.1, alpha = 0.8) +
+   scale_color_manual(values = c("Minor" = "#00602C",
+                                 "Major" = "#31A354",
+                                 "Fatal" = "#A1D99B")) +
+   scale_size_continuous(range = c(5, 15)) +
+   labs(title = "Traffic Accident Analysis: High-Risk Zones",
+        subtitle = "Visualizing Severity, Frequency, and Accident Type",
+        x = "Location Zone",
+        y = "Accident Severity",
+        size = "Accident\nFrequency",
+        shape = "Accident\nType",
+        color = "Severity\nLevel") +
+   theme_minimal() +
+   theme(
+     plot.title = element_text(face = "bold", size = 16),
+     axis.text = element_text(size = 12),
+     legend.position = "right",
+     legend.key.size = unit(0.4, "cm"),
+     legend.text = element_text(size = 8),
+     legend.title = element_text(size = 9, face = "bold"),
+     legend.spacing.y = unit(0.2, "cm")
+   )
>
```

	Location	Accident_Type	Severity	Frequency
1	City Road	Collision	Fatal	4
2	City Road	Collision	Major	4
3	City Road	Collision	Minor	1
4	City Road	Overturn	Fatal	1
5	City Road	Overturn	Major	2
6	City Road	Overturn	Minor	2
7	City Road	Skid	Fatal	2
8	City Road	Skid	Major	2
9	City Road	Skid	Minor	2
10	Highway	Collision	Fatal	1
11	Highway	Collision	Major	1
12	Highway	Collision	Minor	2
13	Highway	Overturn	Fatal	1
14	Highway	Overturn	Major	1
15	Highway	Overturn	Minor	3
16	Highway	Skid	Fatal	3
17	Highway	Skid	Major	2
18	Highway	Skid	Minor	3
19	Junction	Collision	Fatal	2
20	Junction	Collision	Major	2
21	Junction	Collision	Minor	1
22	Junction	Overturn	Fatal	1
23	Junction	Skid	Fatal	2
24	Junction	Skid	Major	3
25	Junction	Skid	Minor	2

CODE4.R* ×data ×

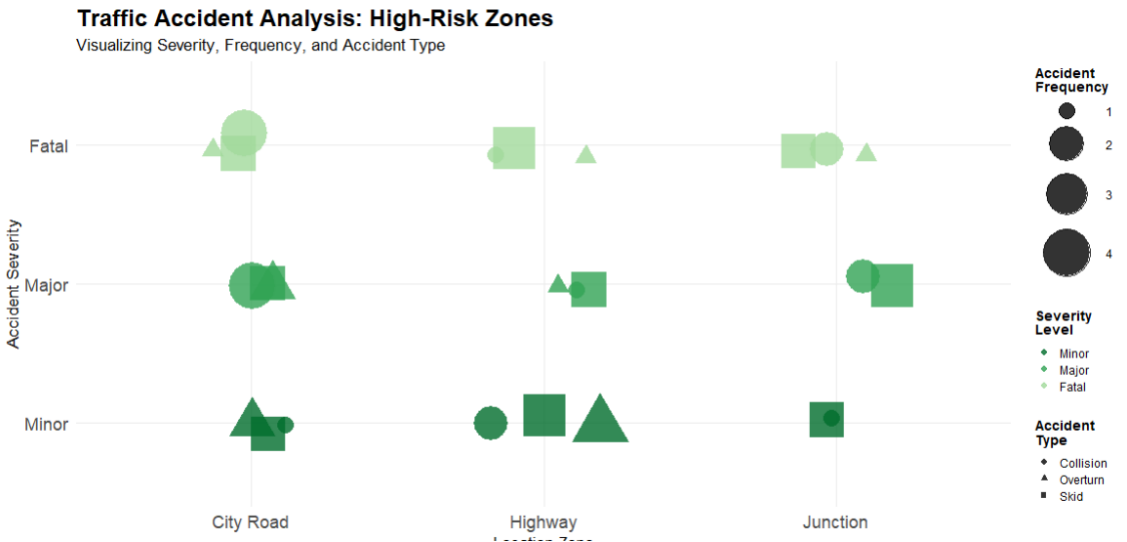
Filter

	Accident_ID	Location	Date	Time	Accident_Type	Severity	Vehicles_Involved	Casualties
1	7001	Highway	2024-06-01	Evening	Overturn	Minor	1	0
2	7002	Highway	2024-06-02	Evening	Skid	Minor	1	1
3	7003	Highway	2024-06-03	Night	Overturn	Fatal	4	0
4	7004	Junction	2024-06-04	Afternoon	Collision	Major	1	3
5	7005	Highway	2024-06-05	Evening	Skid	Minor	1	2
6	7006	Highway	2024-06-06	Afternoon	Overturn	Minor	1	1
7	7007	Highway	2024-06-07	Evening	Collision	Major	1	0
8	7008	City Road	2024-06-08	Morning	Skid	Minor	3	4
9	7009	City Road	2024-06-09	Evening	Collision	Major	4	1
10	7010	Junction	2024-06-10	Morning	Skid	Major	2	1
11	7011	Junction	2024-06-11	Evening	Collision	Major	1	0
12	7012	City Road	2024-06-12	Night	Collision	Major	3	1
13	7013	City Road	2024-06-13	Morning	Overturn	Major	3	4
14	7014	City Road	2024-06-14	Evening	Collision	Fatal	3	5
15	7015	Highway	2024-06-15	Night	Skid	Minor	2	1
16	7016	Highway	2024-06-16	Morning	Skid	Fatal	1	5
17	7017	City Road	2024-06-17	Night	Overturn	Minor	4	4
18	7018	Highway	2024-06-18	Afternoon	Skid	Fatal	4	0
19	7019	Highway	2024-06-19	Afternoon	Skid	Major	3	2
20	7020	Junction	2024-06-20	Morning	Skid	Minor	2	1
21	7021	City Road	2024-06-21	Evening	Collision	Minor	2	2
22	7022	City Road	2024-06-22	Morning	Collision	Fatal	2	4
23	7023	Highway	2024-06-23	Night	Skid	Fatal	4	1
24	7024	City Road	2024-06-24	Evening	Overturn	Minor	3	5
25	7025	Junction	2024-06-25	Afternoon	Collision	Fatal	1	3

FilesPlotsPackagesHelpViewerPresentation

ZoomExport

Publish



POST-LAB QUESTIONS

1. Which visual variable best conveys severity and why?
Color encoding is the most effective way to represent severity because it enables fast and intuitive distinction between categories such as Minor, Major, and Fatal accidents. Human visual perception processes color differences more quickly than variations in shape or size, allowing severity to be recognized instantly across different locations.
2. How can poor encoding mislead policy decisions?
Inappropriate encoding—such as using similar colors for different severity levels or inconsistent scaling—can obscure important trends, including the true impact of fatal accidents. This may cause policymakers to underestimate risks and allocate resources inefficiently.
3. What precautions should be taken for color-blind users?
Designs should use color-blind-friendly palettes with adequate contrast and should not rely solely on color. Supplementing color with additional cues like shapes or spatial positioning enhances accessibility and ensures accurate interpretation.
4. How does visual encoding affect AI explainability?
Effective visual encoding clarifies how patterns and features contribute to outcomes, improving transparency in AI-driven insights. Poorly designed or cluttered visuals can reduce trust and make it difficult to justify AI-based decisions to stakeholders.
5. Suggest improvements for large-scale public dashboards.
Incorporating interactive filters, informative tooltips, scalable legends, and responsive layouts can significantly improve usability. Consistent encoding, simplified visuals, and drill-down capabilities help accommodate a wide range of users and analytical needs.

LEARNING OUTCOME: Students learn perceptually effective visual communication for AI systems.

ASSESSMENT

Description	Max Marks	Marks Awarded
Pre Lab Exercise	5	
In Lab Exercise	10	
Post Lab Exercise	5	
Viva	10	
Total	30	
Faculty Signature		